

Multi-Channel Mechanistic Signatures of Agent WANDERING: Classification, Causal Localization, and Behavior-Legible Rescuability

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Abstract

WANDERING — the failure mode in which an LLM agent stays internally confident it has solved a task yet never emits a termination action and exhausts its turn budget — is a 34% blind spot in probe-based agent monitoring [1]. We characterize WANDERING mechanistically on $N=99$ Qwen3.6-27B SWE-bench Pro trajectories (SUCCESS=40, LOCKED=39, WANDERING=20) by extracting 60 multi-channel features (text, tool-use, per-layer residual at L11/23/31/43/55, and temporal) and fitting a 3-way L1 classifier under nested cross-validation. After removing nine label-leaking features we report a clean macro-F1 of 0.636 ± 0.035 ($z=+5.88$, $p=0.001$ vs. a 1000-permutation null). SUCCESS is robustly separable (AUROC 1.0, driven by tool diversity), but the LOCKED–WANDERING boundary is mechanistically weak ($p=0.035$), echoing a text-based human taxonomy [2] that places 60% of both into a single “Non-Progressive Iteration” category. Stability selection independently surfaces the mid-to-edge mechanism hypothesized in [1]: LOCKED is dominated by L43 mid-layer convergence, WANDERING by L11 edge-layer drift, with effect sizes that are statistically robust but small (2–3 percentage points in cosine). Finally, we ask whether the signature predicts *causal rescuability* — which WANDERING agents recover under L11 SUCCESS-direction injection (paper II). The residual signature does not (LOO-AUC 0.619, $p=0.16$), but tool-entropy collapse depth does (AUC 0.768, $p \approx 0.06$): the more an agent has collapsed onto a single repeated tool, the more rescuable it is. Causal actionability is residual-blind but behavior-legible. We release all code, features, and labels.

1 Introduction

Mechanistic interpretability for *agents* — as opposed to single forward passes — is nascent. Long-horizon LLM agents fail in ways that are invisible to a single-token probe: the model may consolidate an internal “I am done” verdict mid-trajectory yet never translate it into a `finish_tool` action, looping on its action space until it exhausts the turn budget. We call this **WANDERING**, following the operational definition of [1]: a failed trajectory (`finish_reason=max_turns`) whose capability probe stays high (median final score 1.0) — the agent “knows” it has solved the task but never acts on that knowledge.

Companion work [1] showed WANDERING is *detectable* — tool-entropy collapse over the last ten turns separates WANDERING from SUCCESS across three model families (Qwen3.6-27B, Llama-70B, GPT-5) — and proposed a mechanism (§13 therein): the mid-layer verdict is consolidated but an edge-layer (L11/L55) alignment step fails to emit the action. That paper,

however, treated WANDERING as a single phenomenon detected by a single channel. Three questions remain open: (i) Is WANDERING mechanistically *one* thing, or does it decompose? (ii) Does the proposed mid-to-edge mechanism survive an analysis that does not assume it? (iii) Is the signature merely descriptive, or does it predict which agents can be causally *rescued*?

We answer all three on the same $N=99$ corpus, with four contributions:

1. **A multi-channel signature.** 60 features across text, tool-use, residual, and temporal channels classify the three sub-classes (SUCCESS/LOCKED/WANDERING) at macro-F1 0.636 ($p=0.001$). We report a transparent walk-back from a leaky 0.987 baseline.
2. **Independent mechanism confirmation.** Stability selection — using none of [1]’s detector designs — recovers its mid-to-edge picture: LOCKED $\rightarrow$ L43 mid-layer freeze, WANDERING $\rightarrow$ L11 edge-layer drift. Effect sizes are robust but small, and we say so.
3. **A bridge to a human taxonomy.** An LLM-judge mapping to [2] places 60% of both LOCKED and WANDERING into one text category, matching our finding that the LOCKED–WANDERING boundary is mechanistically weak ($p=0.035$).
4. **Behavior-legible rescuability.** Using the causal-rescue outcomes of paper II as labels, we show the residual signature does *not* predict rescuability (LOO-AUC 0.619, $p=0.16$) but tool-entropy collapse depth does (AUC 0.768, $p \approx 0.06$) — the causally actionable signal is behavioral, not residual.

2 Related Work

Agent failure taxonomies. [2] build a 3-phase / 9-category / 20-subcategory taxonomy of SWE-bench failures via human open-coding (Cohen’s $\kappa=0.72$ – 0.77); their “Non-Progressive Iteration” (C2.1) and “Context Amnesia” (C3) overlap conceptually with WANDERING but are not automatically detectable from model internals. [3] characterize success/failure code-agent trajectories with length-distribution statistics (“Quick” vs. “Extended” failures), text-only and without a formal taxonomy. [4] course-correct SWE agents with inference-time process reward models, a *text*-level intervention with no model-internal signal. None use residual-stream features or predict causal rescuability.

Mechanistic interpretability of agent tool use. [5] apply sparse autoencoders and linear probes to pre-action residuals to predict tool-need and tool-risk, but evaluate on function-calling benchmarks rather than long-horizon failure, and do not cluster or causally test failure modes. Our work is, to our knowledge, the first to combine residual-stream features, multi-channel classification, a human-taxonomy bridge, and causal-rescuability prediction on the same agent-trajectory corpus.

Detection of WANDERING. The companion paper [1] establishes the detectors (forensic text $v1$, cross-layer probe disagreement $v4$ with a ~ 15 -turn early-warning lead, tool-entropy $v5$) and their cross-architecture generalization, plus a cross-task null on METR MALT ($p=0.81$) scoping the claim to multi-turn code-execution agents. The present paper takes those sub-class labels as given and asks what the internals say.

3 Data and Methods

Corpus. $N=99$ multi-turn trajectories from Qwen3.6-27B (`enable_thinking=True`, temperature 1.0) on SWE-bench Pro (qutebrowser / openlibrary / ansible), collected on NVIDIA RTX

6000 Pro Blackwell. Per-turn residual activations are captured at L11/23/31/43/55 (bfloat16, $d_{\text{model}}=5120$). The dataset (99 traces + 99 patches + per-turn residual safetensors + derived features) is public:

huggingface.co/datasets/caiovicentino1/swebench-pro-qwen36-27b-phase6

Sub-class derivation. SUCCESS: test passes after `finish_tool` ($n=40$). LOCKED: failed and probe stably < 0.40 ($n=39$). WANDERING: failed and probe stays > 0.70 with no `finish_tool` ($n=20$).

Feature channels (63 raw $\rightarrow$ 60 used). *Text* (11): doubt / exploration / completion word densities, thinking and content lengths, n-gram overlap. *Tool-use* (9): Shannon entropy of tool calls (full / first-10 / last-10), unique-tool count, top-tool fraction, switching rate, longest same-tool run, calls per turn. *Residual* (32): per-layer norm mean/std/late-mean, consecutive-turn cosine (mean and late), first-vs-last drift cosine, plus two cross-layer aggregates. *Temporal* (11): turn count, patch size, probe-trajectory statistics, detector fire-turns.

Label-leakage control. Nine features are removed before the clean analysis: `probe_score_*` (six features derived from the same probe used to bin sub-class via the lock threshold), `n_turns` and `patch_n_bytes` (termination proxies), and `detector_v4_fire_turn` (assignment-process correlate). Two all-NaN detector columns and one zero-variance column are also dropped, leaving 52 features for the primary model.

Pipeline. 3-way L1 multinomial logistic regression (saga), nested 5×3 cross-validation (inner GridSearch over $C \in \{10^{-3}, \dots, 10^2\}$; modal $C=1.0$, interior to the grid), `StandardScaler` inside the pipeline (no test leakage), 1000-permutation null with the Phipson–Smyth $(1 + \#\{\text{null} \geq \text{obs}\}) / (1 + n)$ p -value, and stability selection over 200 bootstraps (feature stable iff selected in $\geq 80\%$, Meinshausen–Bühlmann).

4 Results

4.1 Three-way classification, with a transparent walk-back

We report the leaky baseline alongside the cleaned analysis (per a pre-registered transparency standard):

Run	Macro-F1	z	p
Leaky baseline (uses probe / termination proxies)	0.987 ± 0.030	+12.43	0.001
Clean (primary)	0.636 ± 0.035	+5.88	0.001

The leaky model is near-perfect only because `probe_score_*` is definitionally tied to the sub-class boundary; removing it drops F1 to 0.636, still highly significant against the permutation null (mean 0.322 ± 0.053). Per-class AUROC: SUCCESS 1.000, LOCKED 0.864, WANDERING 0.795. The confusion matrix shows the structure plainly:

true \ pred	LOCKED	SUCCESS	WANDERING
LOCKED	24	0	15
SUCCESS	0	40	0
WANDERING	13	0	7

SUCCESS is perfectly separated; the 28 LOCKED↔WANDERING confusions are the entire error budget.

Pairwise contrasts confirm: WANDERING-vs-SUCCESS and SUCCESS-vs-LOCKED both reach macro-F1 1.000 ($p=0.005$), but **WANDERING-vs-LOCKED is only** 0.631 ($z=1.81$, $p=0.035$) — barely significant. The two failure modes share mechanistic territory.

4.2 Stability selection recovers the mid-to-edge mechanism

Three features are stable (≥ 0.80 selection frequency), one per class:

Feature	Class	Sel. freq	Per-class median (S / L / W)
tool_diversity_count	SUCCESS	1.00	3.0 / 2.0 / 2.0
L43_cosine_consec_late	LOCKED	0.835	0.769 / 0.839 / 0.814
L11_drift_first_last	WANDERING	0.95	0.973 / 0.975 / 0.956

This independently reproduces the [1] §13 picture, using none of its detector designs: **LOCKED** is marked by high L43 mid-layer consecutive-turn cosine (the mid-layer representation *freezes* late — a consolidated wrong verdict that stops updating); **WANDERING** by low L11 first-vs-last drift cosine (the edge-layer representation keeps *rotating* — the model never consolidates to an action); **SUCCESS** by tool diversity (it reaches the third available tool). An ultra-conservative re-fit that drops *all* L43 features and `tool_diversity` still surfaces `L11_drift_first_last` in the top three for the WANDERING-vs-LOCKED contrast (AUC 0.696, $z=1.88$, $p=0.030$), so the edge-layer signal does not depend on the mid-layer or tool features.

Effect sizes are subtle. The discriminating medians differ by only 2–3 percentage points on a 0–1 cosine scale, with overlapping inter-quartile ranges. The signal is a statistical regularity captured by L1 selection across folds, not a visible gap; we make no stronger claim.

4.3 Bridge to a human taxonomy

Mapping all 99 trajectories to [2] via an LLM judge: SUCCESS maps cleanly to a “no-failure” sentinel (39/40), but **60% of both LOCKED (23/39) and WANDERING (12/20) map to the single category C2.1 “Non-Progressive Iteration.”** A text-based taxonomy cannot separate the two sub-types, exactly matching the mechanistic weakness of the LOCKED–WANDERING boundary ($p=0.035$). The probe-derived sub-class definitions may be imposing categorical boundaries on what is, behaviorally and mechanistically, a continuum.

4.4 Behavior-legible rescuability

Paper II shows L11 SUCCESS-direction injection causally rescues 6/20 WANDERING agents (Fisher $p=0.01$ vs. a 0/20 no-hook baseline), with strong per-instance heterogeneity. We ask, *pre-registered*: does the multi-channel signature predict which agents are rescuable, using only features from the original no-hook trajectory? Decision gate: GO iff leave-one-out AUC ≥ 0.70 and permutation $p < 0.10$.

The pre-registered mechanistic model fails the gate (LOO-AUC 0.619, $p=0.16$). The residual channels that classify the sub-classes do not predict causal rescuability. But a single *behavioral* channel does (exploratory, flagged as not the pre-registered primary):

Feature	Univariate AUC	Perm p	Δ median (resc. – resist.)
L11_drift_first_last	0.560	—	−0.010
L11_cosine_consec_late	0.619	—	+0.003
L43_cosine_consec_late	0.500	—	+0.001
tool_entropy_last10	0.768	0.066	−0.487

Rescued agents had much *lower* original tool-entropy (median 0.234 vs. 0.722; Mann–Whitney $p=0.060$): **the more an agent has collapsed onto a single repeated tool, the more rescuable it is by SUCCESS-direction injection.** The SUCCESS direction appears to “unstick” agents trapped in a tight tool-loop, but cannot redirect agents flailing across many tools. Strikingly, the *same* tool-entropy signal that *detects* WANDERING [1] also *selects* which agents are salvageable — the causally actionable information is behavioral, not residual. We treat this as exploratory ($N=20$, 6 events) and flag confirmatory replication as required; it nonetheless points to a tool-entropy-gated detection→intervention loop as the natural next experiment.

4.5 Run-to-run stability prediction

Re-running the 20 WANDERING instances no-hook on the same hardware yields variable outcomes under temperature 1.0 (an independent re-run flips 7/20 to **finish_tool**; a fresh determinism check flips 0/5). Predicting which instances flip from their original features gives LOO-AUC 0.725 ($p=0.096$, marginal), with **L11_cosine_consec_late** the top predictor — a second, independent appearance of an L11 edge-layer signal. This is a run-to-run (temperature-sampling) effect, *not* a hardware effect: all runs used the same GPU.

5 Discussion

WANDERING is a weak boundary, not a clean type. Three independent axes agree: the supervised classifier ($p=0.035$), the human-taxonomy bridge (60% co-located), and the rescuability analysis (heterogeneous) all say LOCKED and WANDERING share territory. The probe-threshold definition draws a line through a continuum. This is a caution for any agent-failure benchmark that treats such labels as ground truth.

The mechanism survives an agnostic analysis — subtly. That stability selection, with no knowledge of [1]’s mid-to-edge hypothesis, recovers L43-mid-for-LOCKED and L11-edge-for-WANDERING is genuine independent corroboration. But the 2–3pp effect sizes mean this is a statistical signature, not a smoking gun; we resist over-claiming a clean mechanistic dissociation.

Causal actionability is behavioral. The most consequential result is the dissociation between *descriptive* and *actionable* signal: the residual signature distinguishes the sub-classes but does not say who can be saved, whereas tool-entropy — a cheap behavioral statistic — does. For a deployed monitor, this is good news: the cheapest channel is the one that matters for intervention, and it streams online.

6 Limitations

(i) Single model (Qwen3.6-27B) and single benchmark (SWE-bench Pro); the multi-channel phenotype is not yet cross-validated off-Qwen. (ii) $N=20$ WANDERING binds every within-WANDERING analysis; rescuability (6 events) and run-stability prediction are underpowered and the rescuability positive is exploratory. (iii) Effect sizes are small (2–3pp cosine). (iv)

L11_drift_first_last correlates with thinking-length ($\rho=0.63$); a partial-correlation control finds L11 retains signal ($\rho_{\text{partial}} = -0.29$, $p=0.004$) but a matched-subset test is marginal ($p=0.08$), so we cannot fully rule out a thinking-volume confound. (v) The detector implementations used for feature extraction differ slightly from [1]’s; two detector-fire columns were all-NaN and dropped.

7 Conclusion

WANDERING is detectable, partially causally rescuable, and mechanistically a weak boundary rather than a clean type. Its residual signature confirms a mid-to-edge picture but does not tell you who to rescue; the behavioral tool-entropy channel does. The detection signal and the rescuability-selection signal are the same cheap statistic — which is what makes a closed detection→intervention loop plausible, and is our next pre-registered target.

Reproducibility. All code, the 60-feature matrix, sub-class and rescue labels, permutation nulls, and the LLM-judge prompt are released under Apache-2.0 at github.com/OpenInterpretability/openinterp-swebench-harness; the dataset (traces + residuals) is on HuggingFace at the link above. Companion papers: Tool-Entropy Collapse (DOI 10.5281/zenodo.20368601); Two Nulls and a Partial Positive (in preparation).

References

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